

Ether Based Electrolyte, $\text{LiB}(\text{CN})_4$ Salt and Binder Degradation in the $\text{Li}-\text{O}_2$ Battery Studied by Hard X-ray Photoelectron Spectroscopy (HAXPES)

Reza Younesi,^{*,†} Maria Hahlin,[†] Marcel Treskow,^{‡,§} Johan Scheers,[‡] Patrik Johansson,^{‡,§} and Kristina Edström^{†,§}

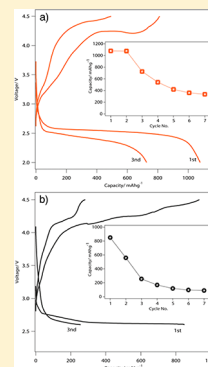
[†]Department of Chemistry—Ångström Laboratory, Uppsala University, Box 538, SE-75121 Uppsala, Sweden

[‡]Department of Applied Physics, Chalmers University of Technology, SE-41296 Göteborg, Sweden

[§]Alistore—European Research Institute, 33 rue Saint-Leu, 80039 Amiens cedex, France

Supporting Information

ABSTRACT: $\text{Li}-\text{O}_2$ cells composed of a carbon cathode containing an $\alpha\text{-MnO}_2$ nanowire catalyst and a Kynar (PVDF-HFP) binder were cycled with different electrolytes containing 0.5 M $\text{LiB}(\text{CN})_4$ salt in polyethylene glycol dimethyl ether (PEGDME) or tetraethylene glycol dimethyl ether (Tetraglyme) solvents. All cells exhibited fast capacity fading. To explain this, the surface chemistry of the carbon electrodes were investigated by synchrotron based hard X-ray photoelectron spectroscopy (HAXPES) using two photon energies of 2300 and 6900 eV. It is shown that the $\text{LiB}(\text{CN})_4$ salt and Kynar binder were degraded during cycling, forming a layer composed of salt and binder residues on the cathode surface. The degradation mechanism of the salt differed in the two tested solvents and, consequently, different types of boron compounds were formed during cycling. Larger amounts of the degraded salt was observed using Tetraglyme as the solvent. With a nonfluorinated Li-salt, the observed formation of LiF , which might be a reason for the observed blockage of pores in the cathode and for the observed capacity fading, must be due to Kynar binder decomposition. The amount of LiF formed in the PEGDME cell was larger than that formed in the Tetraglyme cell. The results indicate that not only the electrolyte solvent, but also electrolyte salt as well as the binder used for the porous cathode must be carefully considered when building a successful rechargeable $\text{Li}-\text{O}_2$ battery.



1. INTRODUCTION

Theoretically, the $\text{Li}-\text{O}_2$ battery has the potential to provide a specific energy about 10 times higher than that of the Li-ion batteries, which is reaching values required to meet the energy storage demand of the future.¹ The practical specific energy of the $\text{Li}-\text{O}_2$ battery is expected to be 2–3 times higher than that of the Li-ion batteries. However, several challenges like rechargeability, safety, cycling efficiency, rate capability, electrolyte stability, and charging overpotential still remain to be solved before we can see a functioning battery on the market.^{2,3}

A $\text{Li}-\text{O}_2$ battery was proposed and already tested in 1996 by Abraham et al.¹ Since then, the rechargeability of the $\text{Li}-\text{O}_2$ battery has been reinvestigated, and it was shown to give a high cycling capacity for roughly 50 cycles.⁴ It was then also suggested that the reaction mechanism consisted of the formation of discharge products of Li_2O_2 or Li_2O that could be split to $\text{O}_2(\text{g})$ and $\text{Li}(\text{s})$ during charging, if an appropriate catalyst was used.⁴ For this, different transition metal oxides have been tested and $\alpha\text{-MnO}_2$ nanowires were suggested as one of the best for a $\text{Li}-\text{O}_2$ battery based on carbonate solvents as electrolytes.⁵

Today a better understanding of the oxygen reduction mechanism has emerged with the oxygen forming a superoxide radical before the reaction with lithium.^{6–8} The superoxide radical has been shown to be highly reactive with the alkyl carbonate based electrolytes like propylene carbonate (PC),

ethylene carbonate (EC), dimethyl carbonate (DMC), etc. by forming carbonate species. The general belief is now that the cycling behavior mainly consists of the cycling of electrolyte reduction products where the $\alpha\text{-MnO}_2$ nanowire catalyst plays an important role for the charging process. Lithium carbonate (Li_2CO_3), lithium alkyl carbonate ($\text{RO}-(\text{C}=\text{O})-\text{OLi}$), Li formate, Li acetate, H_2O , CO_2 , and alkoxides are some of the discharge compounds that have been reported for $\text{Li}-\text{O}_2$ batteries containing alkyl carbonate based electrolytes.^{6–10}

In our previous work, we have in detail studied the interface between the carbon positive electrode and the carbonated based electrolyte in the $\text{Li}-\text{O}_2$ battery.¹¹ The study showed that when using LiPF_6 in PC or in EC/DEC, the reaction product layer on the cathode predominantly contains carbon and oxygen; being approximately 70% of the surface atomic percentage. The remaining part was mainly F and Li, and smaller amounts of P-compounds. The results showed little to no Li_2O_2 and/or Li_2O among the discharge products. It was also observed that the amount of surface fluorine was larger than expected. This could be assigned to the decomposition of the salt^{11,12} but also to instability of the binder. To investigate if these large fluorine

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amounts should be attributed uniquely to the lithium salt or if the binder is also contributing, we here use a nonfluorinated salt, $\text{LiB}(\text{CN})_4$ (LiBison), in two different organic solvents: polyethylene glycol dimethyl ether (PEGDME) and tetraethylene glycol dimethyl ether (Tetraglyme). Thus, for these set-ups, any observed fluorine decomposition products must originate from the binder.

We present here a detailed surface characterization of cycled $\text{Li}-\text{O}_2$ batteries based on the $\text{LiB}(\text{CN})_4$ salt. It has recently been shown that a $\text{LiB}(\text{CN})_4$ -based electrolyte could improve the performance of a $\text{Li}/\text{LiFePO}_4$ cell when compared to a LiBF_4 based electrolyte.¹³ The $\text{B}(\text{CN})_4^-$ anion has a high thermal and electrochemical stability and a weak coordination to the lithium ions. Therefore, it has been suggested as an alternative to PF_6^- .^{14,15} Furthermore, we have chosen to use ether-based solvents: polyethylene glycol dimethyl ether (PEGDME) and tetraethylene glycol dimethyl ether (Tetraglyme), since they have recently been suggested to be stable for $\text{Li}-\text{O}_2$ batteries.^{16,17} There is, however, an ongoing debate whether they are stable against the oxygen reduction reactions.¹⁸ Due to these ambiguities further studies are clearly needed to clarify the limiting reactions in the $\text{Li}-\text{O}_2$ cell with ether based electrolytes. The effect of the salt chemistry and the influence of the cathode components like binder and catalyst composition on cycling behavior must also be properly considered.

Hard X-ray photoelectron spectroscopy (HAXPES) has been used in this study for the investigation of reaction products in the electrode/electrolyte interface as well as of the thickness of the layer. Photoelectron spectroscopy is a powerful technique for the interface analysis since it can provide qualitative and quantitative information about the chemical species on the very thin layers in the order of ~ 5 nm that, in general, is typical for electrode/electrolyte interfaces in lithium batteries. With synchrotron-based PES, the photon energy can be tuned to achieve a nondestructive depth profiling of the reactions products and with HAXPES an enhanced depth of up to 40 nm can be obtained.

2. EXPERIMENTAL SECTION

$\text{LiB}(\text{CN})_4$ was synthesized according to a modified version of the method in ref 13 by using a water suspension of $\text{AgB}(\text{CN})_4$ to which an equimolar amount of LiCl (Merck, 99%) in aqueous solution was added dropwise. The reaction was stirred for 18 h at 80°C in darkness, after which it was filtered and the solvent removed under reduced pressure. The obtained solid was recrystallized from boiling EtOH to remove the majority of unreacted LiCl . The obtained $\text{LiB}(\text{CN})_4$ was dissolved again in a small amount of water and boiled overnight with an additive of $\text{AgB}(\text{CN})_4$ to remove the remaining chlorides. After filtration and evaporation of the solvent, a colorless solid was obtained and finally dried 72 h under vacuum at 70°C . The total yield was 68%. The resulting XRD pattern is provided as Supporting Information (Figure S1). $\text{LiB}(\text{CN})_4$ salt was directly dissolved in polyethylene glycol dimethyl ether, PEGDME, ($M_w = 400$ g/mol, Fluka purum) or tetraethylene glycol dimethyl ether, Tetraglyme, (Merck, >98%)—both dried prior to use—to give ~ 5 mL of 0.5 M electrolytes, and with water content in the electrolytes between 60 to 90 ppm as determined by Karl Fischer titration.

$\text{Li}-\text{O}_2$ cells were assembled within an argon-filled glovebox (O_2 , $\text{H}_2\text{O} < 2$ ppm) using lithium foil as the negative electrode, Solupor as the separator, and a porous cathode film as the positive electrode. The porous cathodes were made of carbon Super P (Lithium battery grade, Erachem Comilog), Kynar 2801

(Arkema) as binder, and $\alpha\text{-MnO}_2$ nanowire as catalyst in weight ratios of 25:33:42. The $\alpha\text{-MnO}_2$ nanowire was synthesized by hydrothermal treating a solution of KMnO_4 and NH_4Cl at 140°C for 24 h as described in ref 19. The cells were connected to a Digatron BTS-600 and cycled in pure oxygen atmosphere at a current density of 70 mA/g of calculated based on the amount of carbon. The cells were stopped at the discharge state limited to 2 and 2.6 V in PEGDME and Tetraglyme cells, respectively, dismantled in the argon glovebox, and then the cathode electrodes were washed with DMC (dimethyl carbonate). The “stored” cathodes had same component as the cycled cathodes but no current or voltage was applied on them. The stored samples were kept in the cell in the oxygen atmosphere for 2 days. Then, the samples were transported to the Helmholtz Centre Berlin (BESSY) synchrotron laboratory in Berlin, Germany, in a vacuum-sealed polymer-coated aluminum bag to protect them against contamination by air or moisture. The bag was opened in an argon glovebox at Bessy, and then the samples were mounted on a specially designed vacuum transfer chamber and transported to the analyzing chamber. The HAXPES measurements were performed on the KMC-1 high kinetic energy photoemission beamline at the HIKE end station at BESSY. To reduce beam intensity and the risk for radiation damage, a $25\ \mu\text{m}$ beryllium filter was used for the measurement with a photon energy of 2300 eV ($\text{Si}(111)$). For the more depth sensitive measurements, a photon energy of 6900 eV was used ($\text{Si}(422)$). The PES measurement on the $\text{LiB}(\text{CN})_4$ salt and the stored samples were performed in-house using a commercial PHI 5500 spectrometer using monochromatized $\text{Al K}\alpha$ radiation and an electron emission angle of 45° . Energy calibration against the hydrocarbon peak at 285.0 eV was applied for all the spectra. The peak intensities were normalized versus number of sweeps, ionization current (reflecting photon flux at the sample during measurements), and the area of the C spectra. Nitrogen gas adsorption experiments were performed at 77 K on a Micromeritics ASAP 2020 surface area and porosity analyzer equipped with the ASAP 2020 V3.04 E software.

3. RESULTS AND DISCUSSION

The results of $\text{Li}-\text{O}_2$ batteries containing nonfluorinated salt and two different ether based solvents will here be described. The characterization has been made with HAXPES to better probe the effect the new $\text{LiB}(\text{CN})_4$ salt and the ether based solvents have on the cycling behavior of the battery.

3.1. Basic Characterization of the Carbon Cathode and Cycling of the Battery. The porous structure carbon cathodes including $\alpha\text{-MnO}_2$ as a catalyst were used to make $\text{Li}-\text{O}_2$ cells. The BET surface area and pore volume of the porous cathode were equal to $37.7\ (\text{m}^2/\text{g})$ and $0.22\ (\text{cm}^3/\text{g})$, respectively. In Figure 1, the discharge/charge characteristics for the assembled cells using 0.5 M $\text{LiB}(\text{CN})_4$ in PEGDME and Tetraglyme solvents, hereafter referred to as PEGDME and Tetraglyme cells, respectively, are shown. The two batteries perform differently with the discharge plateau of the first cycle about 0.5–1 V lower for the PEGDME cell compared to that of the Tetraglyme cell. The charge plateau is above 4.1 V for both cells—which is dramatically higher than for more classical “ LiPF_6 + carbonate based solvent” cells where the charge and discharge plateaus are closer to 3.6–3.8 V and 2.6–2.8 V, respectively.^{17,20} This 0.5–1 V larger overpotential in the case of ether-based solvents could be attributed to a higher stability of these electrolytes, and may also explain the absence of the CO_2 evolution reported for the use of carbonate-based electrolytes.^{6,21,22}

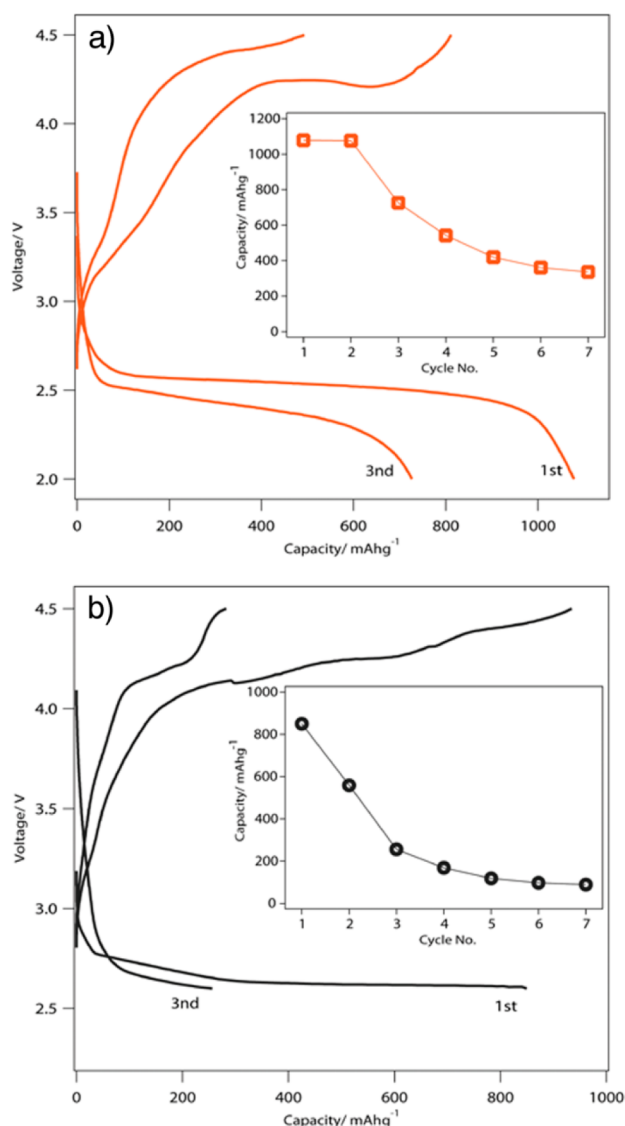


Figure 1. First and third discharge and charge plateaus vs. voltage of the cells with LiB(CN)₄ in PEGDME (a) and Tetraglyme (b) electrolytes, at a current density of 70 mA/g of carbon, respectively. The lower cutoff voltage of the PEGDME and Tetraglyme cells were limited to 2 and 2.6 V, respectively. Inset: Discharge capacities vs cycle number of the cells.

The low cutoff voltage was limited to 2 V in the PEGDME cell and to 2.6 V in the Tetraglyme cell to obtain a discharge capacity close to 1000 mAh/g for the first discharge and prevent fully blockage of cathode pores. Both cells were charged to 4.5 V, since the charge plateau is close to 4.2–4.3 V. After the first cycles, both cells show rapid discharge capacity fading (inset of Figure 1). From first to fourth cycle, the discharge capacity decreased about 50% and 80% in the PEGDME and Tetraglyme cells, respectively.

3.2. Reaction Product Studies Using HAXPES. To investigate the reactions products, the cell cathodes at their respective discharged state were used for HAXPES measurements. The survey spectra showed the presence of F, O, Mn, C, N, B, and Li at the surface layer of the cathodes. The high-resolution F 1s spectra of the reference and stored samples (Figure 2-a) show one clear F 1s peak which is assigned to the F–C bond of the Kynar binder.²³ The F 1s spectra of the cycled samples (Figure 2-a) show two clear peak structures. Here, the

peak at higher binding energy (688.1 eV) originates from the F–C bond of the Kynar binder, while the peak at lower binding energy (685 eV) originates from a compound formed during the cycling of the Li–O₂ cell.

Since there was no fluorine source in the two electrolytes, the new peak at 685 eV clearly represents degradation of the Kynar binder during cell cycling. The binding energy of the formed peak matches the reported binding energy of LiF.^{23,24} Therefore, these results show that fluorine from the binder is partially reacted with lithium from the electrolyte. Importantly, the “stored” cathodes do not show any LiF peak, and this shows that the reduction or oxidation reactions of the Li–O₂ cell facilitate the binder decomposition. As shown in Figure 2a, the relative amount of LiF formed in the PEGDME cell is higher than that in the Tetraglyme cell, showing a higher degree of binder decomposition when using the PEGDME.

LiF is one of the typical components of the SEI layer in lithium ion batteries with electrolytes containing fluorinated salts such as LiPF₆.^{23–25} However, the formation of LiF which insulates carbon and/or catalyst from lithium ion transport and electron transfer is not favorable in the Li–O₂ cell. Binder degradation and LiF formation might, in addition, clog the pores within the cathode which have a critical role to render space for the desired reaction product Li₂O₂ and to increase the discharge capacity.^{26,27} The chosen system in this study, in which the electrolytes do not contain fluorine, facilitate the study of possible degradation of Kynar (PVDF-HFP) used as binder in the porous cathode. Kynar has been employed often to assemble porous air electrodes^{4–7,17,20,28} or polymer electrolyte^{29,30} for Li–O₂ batteries, but any instability in such a system has not earlier been reported. However, for Li-ion batteries, PVDF reduction^{31,32} and PVDF decomposition at high temperature^{33,34} have been previously reported, while the decomposition of PVDF-based binder used in the Li–O₂ cell at room temperature is, to our knowledge, presented here for the first time. Finally, it should be mentioned that CO₂ evolution, which has been reported as an electrolyte decomposition product in the Li–O₂ cell^{6,21,22} might also originate from the degradation of the binder.

All binding energies for the different compounds used for the deconvolution of the measured spectra are displayed in Table 1. The C 1s spectra of the reference, stored, and cycled cathodes and of the LiB(CN)₄ salt are presented in Figure 2-b. The peak at 284.6 eV, which is clearly visible in the reference electrode, represents the super P carbon in cathodes.³⁵ This peak is not visible in the spectra of the PEGDME cell, confirming that a thick layer is deposited on the cathode surface. However, the super P peak in the spectra of the Tetraglyme cell is to some extent visible. There are two other main peaks in the C 1s spectra of the reference electrode at 286.6 and 290.9 eV displaying the carbons in the Kynar binder.^{36,37} These peaks are attributed to the –CH₂– and –CF₂– bonds, respectively, which have an almost equal intensity ratio. However, the intensity ratio of the –CH₂– to –CF₂– increases in the cycled cathodes. This can be a result of binder degradation, but formation of ether-based compounds at 286.5 eV during cycling may partly explain the change in the intensity ratio.³⁸ The remaining LiB(CN)₄ salt on the carbon surface also contributes to the peak at 286.5 eV in the C 1s spectra (Figure 2), as supported by the N 1s and B 1s spectra (Figure 3) which indicate the presence of the salt in the surface layer.

The B 1s and N 1s spectra (Figure 3) show the presence of boron and nitrogen from the LiB(CN)₄ salt in the cycled

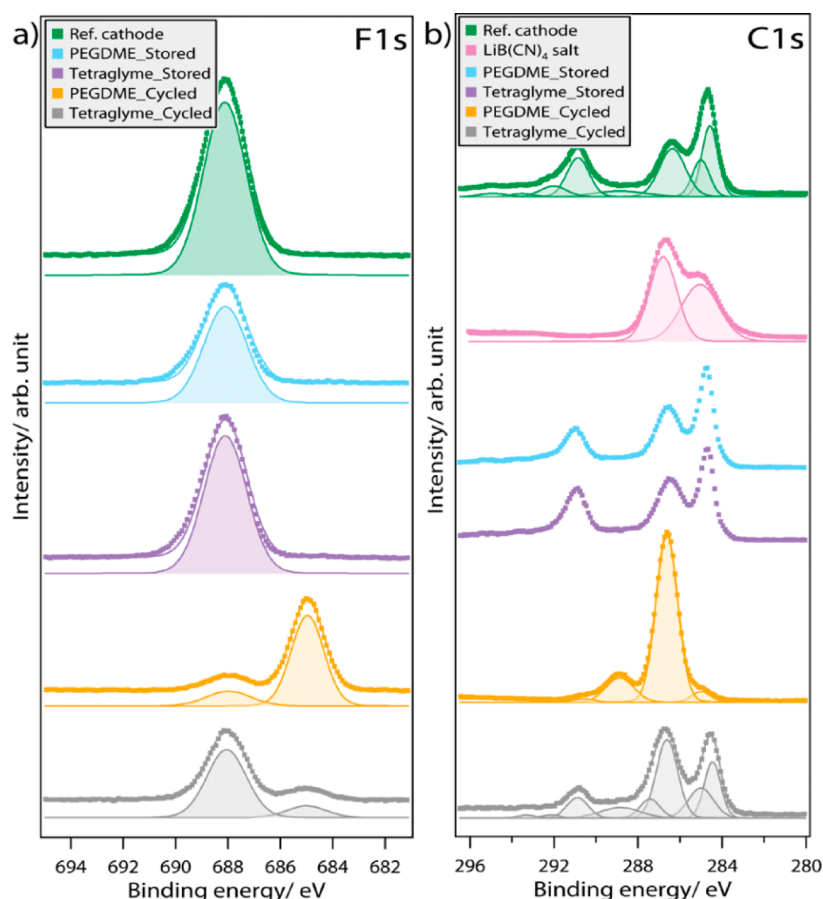


Figure 2. F 1s (a) and C 1s (b) spectra of the reference and cycled cathodes using $\text{LiB}(\text{CN})_4$ in PEGDME or Tetraglyme. The photon energy was 2300 eV.

Table 1. Main XPS Peak Assignments in This Study

assigned compound	measured binding energy (eV)						
	C	F	Mn	O	Li	N	B
super P	284.6						
hydrocarbon	285						
CH_2	286.6	688.1					
CF	288.7						
CF_2 in PVDF	291						
CF_2 in HFP	291.9						
LiF		685			55.3		
MnO_2			642.2, 653.9	529.9			
$\text{LiB}(\text{CN})_4$	286.6					400.1	191.8

cathodes. The N 1s spectra (Figure 3a) of the cycled cathodes all show one peak at a binding energy of 400.1 eV: this represents the N 1s spectra of the $\text{LiB}(\text{CN})_4$ salt powder (Figure 3a).³⁹ The B 1s spectra (Figure 3b) of the cycled cathodes display two peaks originating from different types of boron bonds. The peak at 191.8 eV, which is present in the B 1s spectra of salt and both cycled cathodes, is assigned to the unique B–C bonds in the $\text{LiB}(\text{CN})_4$ salt (Figure 4).³⁹ Besides the salt peak, the PEGDME and Tetraglyme cells show one additional B 1s peak located at a binding energy of 190.1 and 192.7 eV, respectively.

These results clearly show that some of the bonds between B and C in the salt are broken during cycling, forming new types of B-bonds. The peak at 192.7 eV in the Tetraglyme cell shows that B is bonded to an atom more electronegative than the C in the salt, e.g., B–F or B–O bonds.⁴⁰ It has recently been shown that

the degradation of the boron containing LiBOB salt in a Li–O₂ cell leads to the formation of B_2O_3 or LiB_3O_5 .²⁸ Similar reactions could also be present in this system since boron containing $\text{LiB}(\text{CN})_4$ salt is used. The binding energy of the formed peak at 192.7 eV matches the reported binding energy of B_2O_3 .⁴¹ In the case of the PEGDME cell, the additional peak at 190.1 eV suggests that B is bonded to an atom less electronegative than the C in the salt. At present, we have no clear indication of the corresponding species. Altogether, the present results show $\text{LiB}(\text{CN})_4$ salt decomposition during cycling with an additional solvent chemistry influence on the decomposition mechanism.⁴² Moreover, no additional peaks in either the N 1s or the B 1s spectra of the stored samples could be observed, thus indicating that the observed effects are uniquely electrochemical in nature.

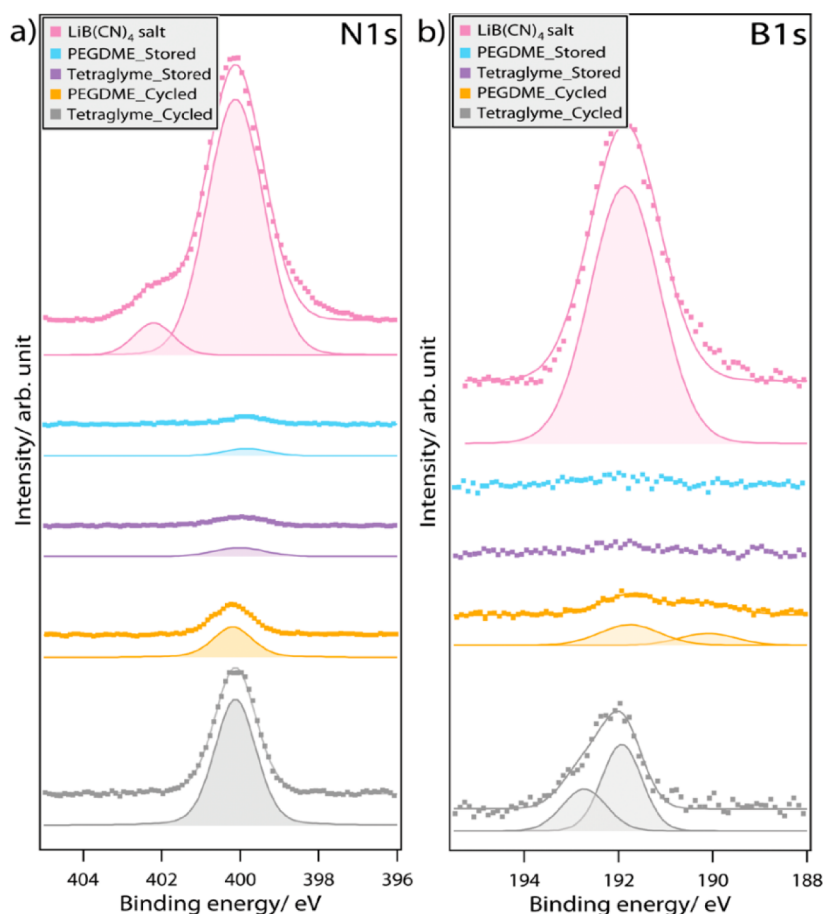


Figure 3. N 1s (a) and B 1s (b) spectra of the cycled cathodes using $\text{LiB}(\text{CN})_4$ in PEGDME or Tetraglyme. The photon energy was 2300 eV.

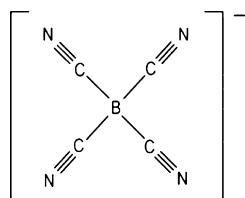


Figure 4. Chemical structure of the $\text{B}(\text{CN})_4^-$ anion.¹³

Figure 5a–c shows the Mn 2p, O 1s, and Li 1s spectra of the reference and cycled cathodes. The Mn 2p spectra of the reference cathode consists of spin–orbit split peaks (Mn 2p_{3/2} and Mn 2p_{1/2}) located around a binding energy of 642.3 and 654 eV, respectively. These two peaks originate from the $\alpha\text{-MnO}_2$ catalyst and show that the oxidation state of manganese is dominantly 4+.⁴³ Although $\alpha\text{-MnO}_2$ catalyst has been commonly used as the catalyst in the Li–O₂ battery, its catalytic activity to decompose Li_2O_2 is questionable. It has been shown that catalysts may rather facilitate the electrolyte decomposition, while not actually catalyze Li_2O_2 decomposition.²² The peak at 530 eV in the O 1s spectra is originating from oxygen in MnO_2 . As shown in the Mn 2p and O 1s spectra, the MnO_2 peaks are clearly present in the spectra of the reference cathode, while not in the spectra of the PEGDME cell. The absence of catalyst peaks in the Mn 2p and O 1s spectra of the PEGDME cell reveal that a thick layer is formed on top of the catalyst nanowires. This is in agreement with earlier conclusions from the F 1s and C 1s spectra (above). Since the Mn 2p, F 1s, and C 1s spectra were obtained

by using a photon energy of 2300 eV, it could be concluded that the deposited passive layer had a thickness >2 nm.^{44,45}

Furthermore, the catalyst peaks are to some extent visible in the Mn 2p and O 1s spectra of Tetraglyme cell. Hence, it is concluded that a thinner layer is formed in the Tetraglyme cell compared to the PEGDME cell. Also, this is in agreement with the F 1s spectra—showing less LiF formation on the cathode of the Tetraglyme cell. The Li 1s spectrum of PEGDME cell displays a high intensity peak at 55.3 eV, while the Tetraglyme cell has a small intensity peak at 55.5 eV. These peaks could both be assigned to the formed LiF seen in the F 1s spectra.⁴⁶ However, it has been reported that ether-based solvents decomposition products include Li_2CO_3 , HCO_2Li , and $\text{CH}_3\text{CO}_2\text{Li}$.¹⁸ Therefore, the Li 1s peaks might also originate from these products.

3.3. Depth-Profiling. Spectra were also recorded using a higher photon energy (6900 eV) to obtain a depth-profile of the deposited layer at the discharged states (Figure 6a–d). Changing the photon energy from 2300 eV to a photon energy of 6900 eV implies that the mean free path of electrons changes from approximately 5 to 20 nm,^{44,45} and that a more bulk sensitive information is gained. The spectra obtained using the different photon energies are very similar to each other, confirming the presence of same type of compounds also deeper into the discharge product layer. A detailed curve fitting of the F 1s, O 1s, N 1s, and C 1s spectra of the cathodes obtained at 6900 eV is presented in the Supporting Information (Figure S2).

The F 1s, O 1s, N 1s and C 1s spectra of the cathodes of the PEGDME and Tetraglyme cells measured with the two different

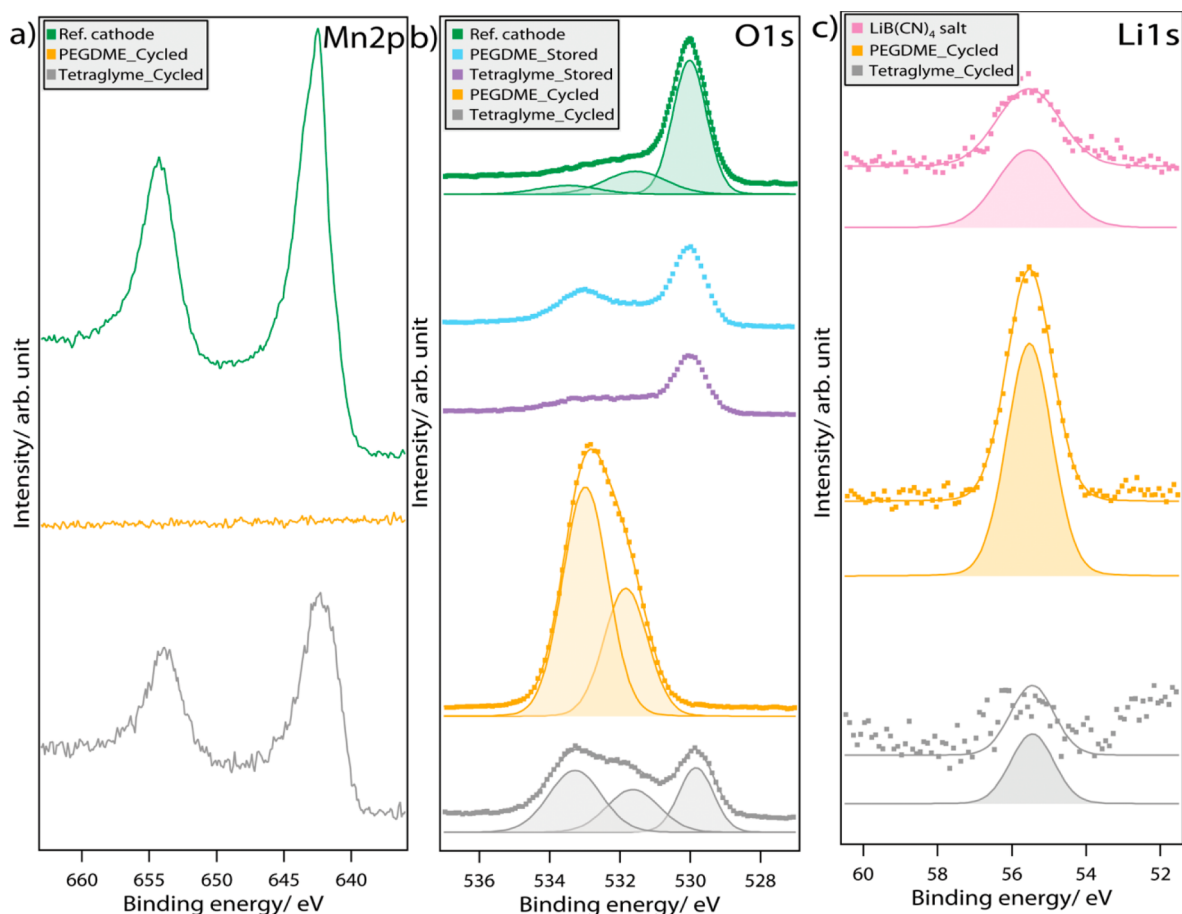


Figure 5. Mn 2p (a), O 1s (b), and Li 1s (c) spectra of the reference and cycled cathodes using $\text{LiB}(\text{CN})_4$ in PEGDME or Tetraglyme. The photon energy was 2300 eV.

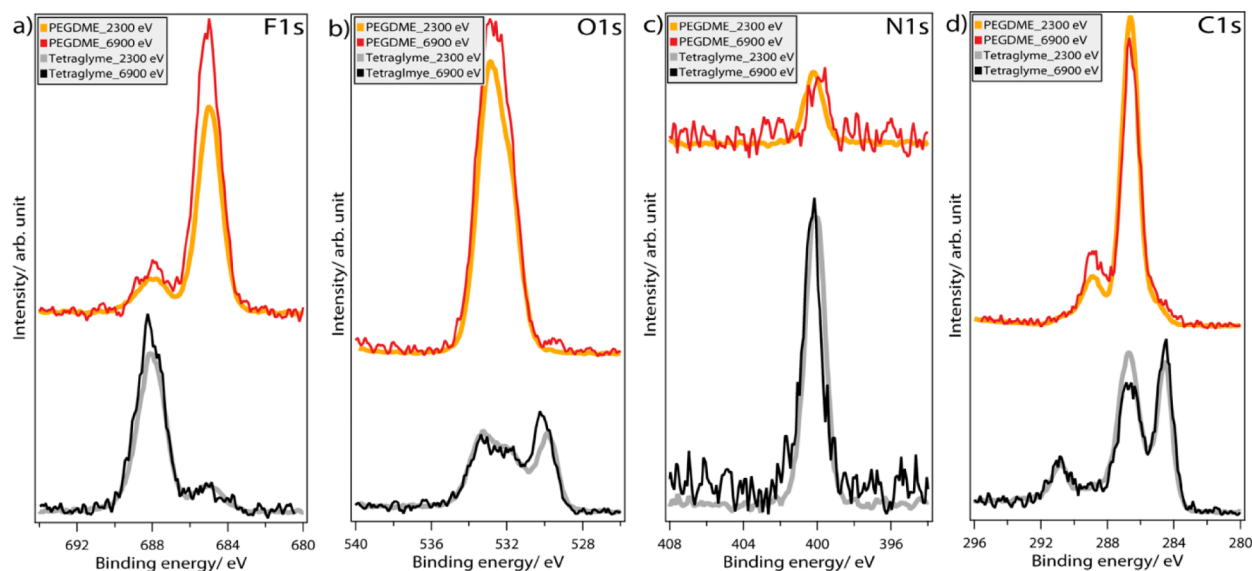


Figure 6. F 1s (a), O 1s (b), N 1s (c), and C 1s (d) spectra of the cycled cathodes using $\text{LiB}(\text{CN})_4$ in PEGDME (orange and red curves) or Tetraglyme (gray and black curves). The photon energies were 2300 and 6900 eV.

photon energies are compared in Figure 6. The peak at the binding energy of 658 eV in the spectra of both PEGDME and Tetraglyme cells show that the binder decomposition product, LiF, is present also at larger depth. The relative intensity of the LiF peak to the binder peak for the PEGDME cell increases at

larger depth showing that more binder decomposition products are present closer to the electrode surface.

The MnO_2 peak in the O 1s spectra (Figure 6b) of PEGDME cell becomes visible when using a higher photon energy. This shows the presence of MnO_2 catalyst at the depth of the surface

layer on the cathode, and indicate that the thickness of the deposited layer in the cathode of PEGDME cell is thinner than 20 nm. For both PEGDME and Tetraglyme cells, the relative intensities of both N 1s peaks (Figure 6c) are the same when using both photon energies, and therefore, the results show that the salt is evenly distributed in the formed discharge layer on the cathode surface.

Overall, the results indicate that degradations of salts and binders as well as of the solvents need to be considered comprehensively for designing a functioning rechargeable Li–O₂ cell. Kynar binder decomposition forms a layer of LiF on the surface of cathode. Binder free cathode design could be a solution to avoid LiF formation.⁴⁷ Other types of binder like poly acrylic acid (PAA) or carboxymethyl cellulose (CMC) might be more stable in the Li–O₂ battery.

4. CONCLUSIONS

The Li–O₂ batteries cycled using the nonfluorinated LiB(CN)₄ salt and PEGDME or Tetraglyme solvents show a rapid capacity fading. The HAXPES study clearly shows that the LiB(CN)₄ salt degraded during cycling and more so in the Tetraglyme cell compared to the PEGDME cell. Moreover, peaks with different binding energies in the B 1s spectra reveal that the salt decomposition mechanism depends on the solvent chemistry. By using a nonfluorinated salt, it is here unambiguously shown that the Kynar (PVDF-HFP) binder also decomposes, being the sole fluorine source to the observed formation of LiF. The binder decomposition was observed to be larger in the PEGDME cell compared to that of the Tetraglyme cell. The PES results using photon energies of 2300 and 6900 indicate that the deposited layer is >5 nm and <20 nm in the PEGDME cell, while <2 nm in the Tetraglyme cell. Altogether, the results show that not only the stability of the salt and solvent(s) in the Li–O₂ battery needs to be considered, but also the stability of the binder.

■ ASSOCIATED CONTENT

■ Supporting Information

XRD pattern of synthesized LiB(CN)₄ salt. The F 1s (a), O 1s (b), N 1s (c), and C 1s (d) spectra of the cycled cathodes using LiB(CN)₄ in PEGDME or Tetraglyme measured with the photon energy of 6900 eV. This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*Tel: +46 18 471 3769; e-mail: reza.younesi@kemi.uu.se.

Notes

The authors declare no competing financial interest.

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